

- Q.27 Explain the working principle of opto-isolator.
- Q.28 What is Light emitting diode?
- Q.29 What is linear velocity and explain any one method to measure the linear velocity?
- Q.30 What is Doppler transducer and write its two applications?
- Q.31 What is the strain gauge Accelerometer.
- Q.32 What are the limitation of ultrasonic methods.
- Q.33 Discuss about measurement of thickness and explain any one method.
- Q.34 Describe a method to measure the viscosity.
- Q.35 How can we measure the pH?

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is LVDT? Explain in detail
- Q.37 Write a short note on any two:
- Protector
 - Verneir calliper
 - Clinometer
- Q.38 Explain Techogenerator in detail? Give its applications.

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6th Sem / IC

Subject:- Advanced Measurement Techniques

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The Angular measuring instrument is/are
- Potentiometer
 - LVDT
 - RVDT
 - All of these
- Q.2 Shaft encoder is an example of
- Analog transducer
 - Active transducer
 - Inverse transducer
 - Digital Transducer
- Q.3 Force = ?
- Mass/Acceleration
 - Acceleration / Mass
 - Mass* Acceleration
 - Mass / Velocity
- Q.4 LED works in_____
- Forward bias
 - Breakdown Region
 - Reverse bias
 - All of the above
- Q.5 In a measuring system quantity under measurement is termed as _____
- Measurand
 - Controllers
 - Sensors
 - Indicators

Q.6 Which of the following element is not part of measuring system

- a) Sensors
- b) Variable conversion elements
- c) Signal processing elements
- d) Human

Q.7 Which transducer is known as 'self-generating transducer'?

- a) Active transducer b) Passive transducer
- c) Secondary transducer d) Analog transducer

Q.8 LVDT works on the principle of

- a) Electromagnetic induction
- b) Mutual induction
- c) Seebeck effect
- d) None of the above

Q.9 Strain gauge is a _____

- a) Resistive transducer
- b) Capacitive transducer
- c) Inductive transducer
- d) Digital transducer

Q.10 Unit of Frequency

- a) Hertz b) Farad
- c) Henry d) Ohm

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Deflecting torque in an instrument may be produced by Electrostatically (T/F)

Q.12 Explain photo electric in brief

Q.13 Explain SCR

Q.14 Photo voltaic cell is inverse Transducer (T/F)

Q.15 Expand LDR

Q.16 Give two example of Data presentation unit.

Q.17 Write the formula of Linear velocity.

Q.18 Unit of Density.

Q.19 Define Specific gravity in brief.

Q.20 What is PH of De-minralized water?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Differentiate between graphical method and numerical method

Q.22 Write a short note on Light suppression.

Q.23 Write a short note on Scale.

Q.24 Explain the dial bevel protector and why it is used?

Q.25 Draw the functional elements of measuring system

Q.26 Describe I/O configuration of instrumentation system